Carbon Quantum Dots in UV/H_2O_2 Dye Treatment

How does adding CQDs or N-CQDs affect the breakdown of methylene blue by H_2O_2 and UV light?

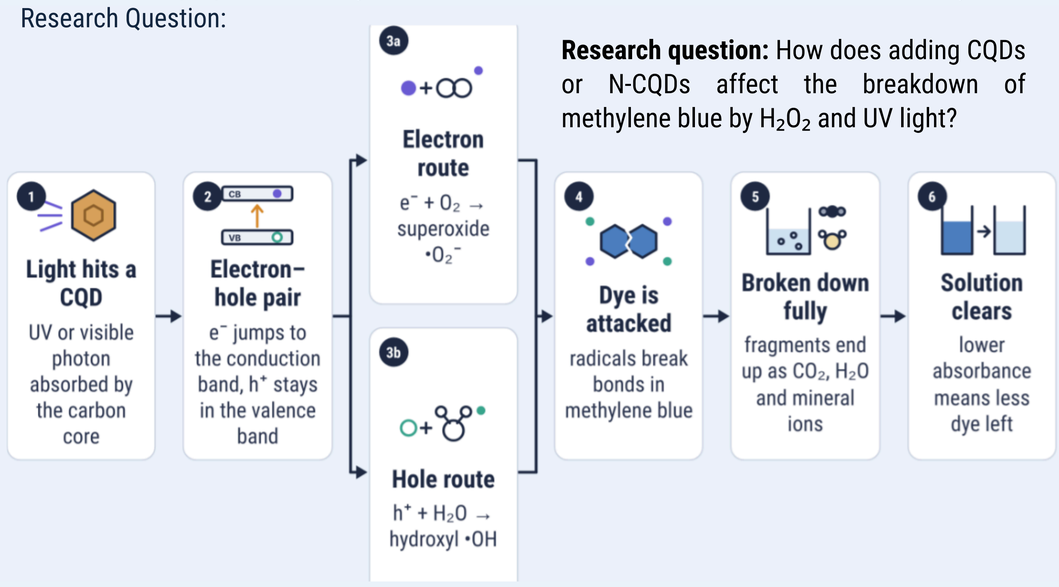
JOHN MONASH SCIENCE SCHOOL

1 Background

Textile and printing industries discharge methylene blue (MB) dye, blocking sunlight in waterways and harming aquatic organisms (Khan et al., 2022). MB is hard to remove with normal treatment, so UV light and hydrogen peroxide (H_2O_2) are used to produce reactive radicals that break it down (Glaze et al., 1987). Carbon quantum dots (CQDs) are carbon particles smaller than 10 nm, made cheaply from citric acid in a microwave. They are reported to act as photocatalysts, accelerating a chemical reaction in light, by absorbing UV and forming additional radicals (Lim et al., 2015; Wang et al., 2017).Adding urea makes nitrogen-doped CQDs (N-CQDs) (Qu et al., 2012), and doping is reported to improve dye breakdown (Yashwanth et al., 2020).

If effective, CQDs could offer a low-cost, non-lethal way to improve dye treatment. Absorbance is proportional to dye concentration, so a lower absorbance indicates less dye.
Research Question:

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2 Carbon Quantum Dots Synthesis & execution

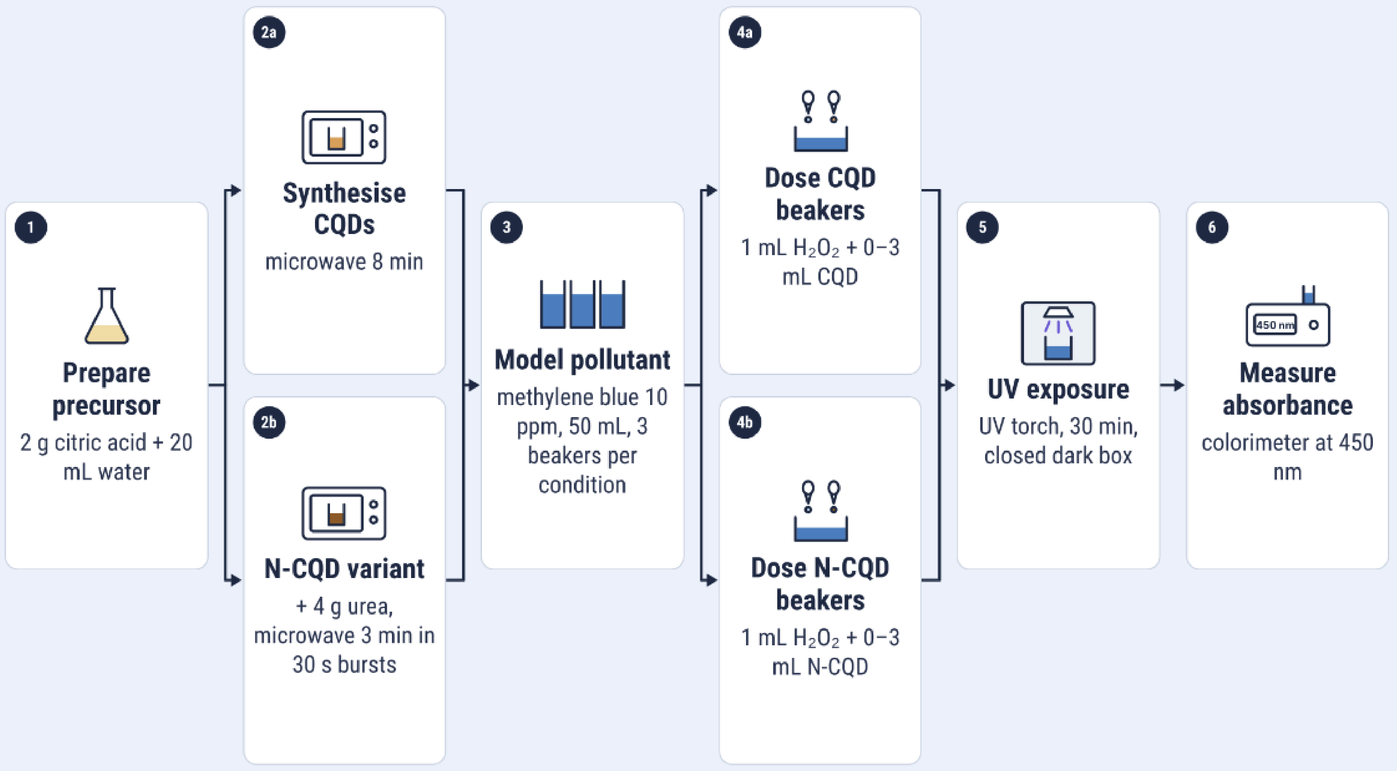


Figure 1. Experimental procedure: CQD and N-CQD synthesis, dosing methylene blue with H_2O_2 and CQDs, UV light exposure, and absorbance measurement at 450nm.

3 Characterisation of Carbon Quantum Dots

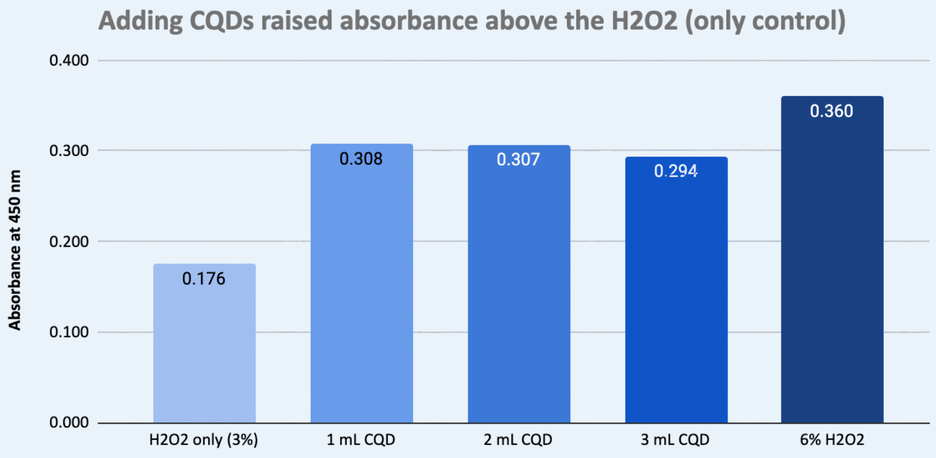


Figure 2. Absorbance after 30 min UV. Control and 6% H_2O_2: mean of 3 trials (bar = range); CQD: 1 trial each.

All CQD samples (0.294–0.308) and the 6% H_2O_2 trial (0.360) read higher than the control (0.176), the opposite of the expected result.

CQD Samples



Figure 4. N-CQD (left) and CQD (right) solutions; both are strongly coloured.



Figure 5. Samples fluorescing green under UV light.

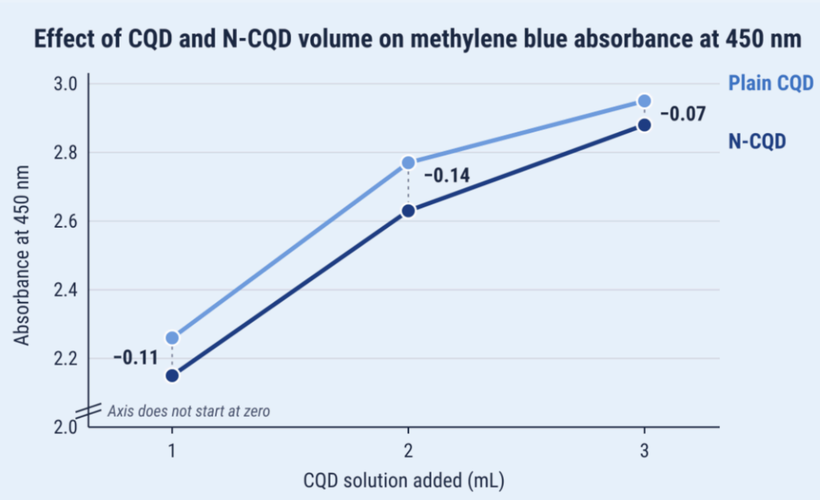


Figure 3. CQD vs N-CQD absorbance by volume (±0.01); separate day, not comparable with Figure 2.

Absorbance rose with volume for both types. N-CQDs read 0.07–0.14 lower than CQDs at every volume.

4 Discussion

All CQD samples (0.294–0.308) were above the 3% H_2O_2 control (0.176). They also read below the matched 6% H_2O_2 control (0.360), despite the CQDs' own absorbance at 450 nm. With one trial per CQD condition and a control range of 0.06, neither difference is conclusive.
A major limitation was wavelength choice. MB absorbs most strongly near 664 nm and transmits blue light, whereas the yellow-brown CQDs absorb strongly at 450 nm. The readings therefore reflected CQD colour more than the dye that remained.
The measurements were also inconsistent. Figure 2 showed no trend with CQD volume, while Figure 3 rose sharply and read 7–10 times higher for similar mixtures, which suggests the colorimeter was calibrated differently on each day. Above an absorbance of 2, less than 1% of light reaches the detector, so the N-CQD/CQD gap is likely within instrument noise.
Raising H_2O_2 from 3% to 6% unexpectedly doubled absorbance, suggesting excess H_2O_2 may have consumed •OH itself (H_2O_2 + •OH → HO_2• + H_2O).
The CQDs may also have slowed the reaction. They absorb UV strongly, which could shield the H_2O_2, and they release some of that energy as fluorescence rather than radicals. Handheld UV torches (365–395 nm) also split H_2O_2 only weakly.

Limitation	Improvement
450 nm overlaps CQD absorption	Measure at 664 nm(the MB absorption peak)
No CQD-only reference	Blank each sample against the same volume of CQD in water
No initial absorbance or calibration	Record before and after readings; apply a 0–10 ppm calibration curve
Adsorption not separated from photocatalysis	Add a dark control (CQD, no UV)
Few replicates; control range 0.144–0.203	Use at least 5 replicates per condition
UV light source providing no detail regarding intensity	Use an alternative UV light source that allows us to identify the exposure as exposure may have been insufficient

5 Future directions and Implementations

Future research

1. Identifying the active species. Radical scavengers, such as isopropanol for •OH and benzoquinone for superoxide, could reveal which radicals are mainly responsible for breaking down the dye.
2. CQD composites. Pairing CQDs with semiconductors such as TiO_2 may improve charge separation and boost photocatalytic activity (Wang et al., 2017).
3. Synthesis and doping. Adjusting microwave time, urea ratio or co-dopants could show which CQD properties make them most effective (Yashwanth et al., 2020).

Potential applications

- Low-cost wastewater treatment using inexpensive, low-toxicity precursors (Qu et al., 2012)
- Sunlight-driven purification, as CQDs can absorb visible light (Wang et al., 2017)
- Pollutant sensing using CQD fluorescence (Figure 5; Lim et al., 2015)
- Safer bioimaging, as fluorescent, low-toxicity CQDs offer an alternative to heavy-metal quantum dots (Lim et al., 2015)

6 Conclusion

- The effect of CQDs and N-CQDs on UV/H_2O_2 degradation of methylene blue could not be determined under the conditions tested.
- CQD samples (0.294–0.308) read above the 3% H_2O_2 control (0.176) but below the matched 6% control (0.360). The N-CQD/CQD difference fell within instrument noise.
- CQD absorption at 450 nm most likely masked any dye removal.
- Measurements at 664 nm with CQD-only blanks and matched controls are required before CQDs can be evaluated as a low-cost dye treatment.

7 References

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